FIN2020 Homework 6

Problem 1

Suppose the expected return on the tangent portfolio is 10% and its volatility is 40%. The risk-free rate is 2%.

(a) What is the equation of the Capital Market Line (CML)?

(b) What is the standard deviation of an efficient portfolio whose expected return of 8%? How would you allocate $1,000 to achieve this position?

Problem 2

Suppose the market premium is 9%, market volatility is 30% and the risk-free rate is 3%.

(a) What is the equation of the SML?

(b) Suppose a security has a beta of 0.6. According to the CAPM, what is its expected return?

(c) A security has a volatility of 60% and a correlation with the market portfolio of 25%. According to the CAPM, what is its expected return?

(d) A security has a volatility of 80% and a correlation with the market portfolio of -25%. According to the CAPM, what is its expected return?

Problem 3

Here are some stocks with high and low betas, as reported by the *Value Line Investment Survey*:

High Beta Stocks		Low Beta Stocks	
Company	Beta	Company	Beta
Intel	1.05	Procter & Gamble	0.65
Ford Motor	1.25	Wal-Mart	0.70
Delta Airlines	1.30	AT&T	0.75
US Steel	1.90	Pfizer	0.90

Suppose that the risk-free rate is $r_f = 0.02$ (2 percent) and that the expected return on the market portfolio is $E(\tilde{r}_M) = 0.08$ (8 percent), then use the CAPM to calculate the expected returns on each of these eight stocks.